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Faculty of Artificial Intelligence and Cybersecurity (FAIX)
BAXU 2213 — SOFTWARE ENGINEERING

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)
for
LOST AND FOUND MANAGEMENT SYSTEM
Version 1.0 (Approved)

Group Mini-Project (20%)
SRS, UML, Project Planning, Testing, and Cost Estimation

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Date Submitted: 13 June 2026
Course Learning Outcome: CLO3 (P3, PLO6, LOD-C3F)

Revision History

Name	Date	Reason For Changes	Version
LFM Team	13/06/2026	Initial version of the SRS document	1.0

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1. Introduction

This section provides an overview of the document and orients the reader to the system being specified. It summarizes the SRS's purpose, product scope, intended audience, and how the document is organized.

1.1 Document Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for Version 1.0 of the Lost and Found Management System. The purpose of this document is to provide a clear and comprehensive description of the system requirements for developers, testers, project managers, and stakeholders throughout the software development lifecycle. This SRS specifies what the system must do, not how it will do it, and serves as the binding foundation for system design, implementation, verification, and maintenance activities.

1.2 Document Conventions

This document follows the IEEE Software Requirements Specification (SRS) standard. All requirements are assigned unique identifiers and are written using clear, concise, and testable statements. The keyword “shall” is used to indicate mandatory requirements, while priorities are categorized as High, Medium, or Low based on their importance. Each requirement includes supporting information such as rationale, acceptance criteria, and verification methods where applicable.

1.3 Intended Audience and Reading Suggestions

This document is intended for multiple primary audiences across the software lifecycle.

- **Developers and System Analysts:** Should focus on Section 2 (Overall Description) for system context and Section 3 (Requirements) for detailed functional and non-functional requirements.
- **Testers and QA Teams:** Should review Section 3 together with Section 4 (Verification) to develop test cases and validate system requirements.
- **Project Managers and Stakeholders:** Should focus on Section 1 (Introduction) and Section 2 (Overall Description) for a high-level understanding of the system scope, objectives, and constraints.

The document is organized sequentially. Readers seeking a general overview should begin with Sections 1 and 2, while technical personnel should proceed to Sections 3 and 4 for detailed specifications and verification information.

1.4 Product Scope

The Lost and Found Management System is a self-contained, web-based application designed to simplify and automate the reporting, tracking, and recovery of lost and found belongings within public environments such as universities, shopping malls, offices, and other public facilities. The system aims to improve the efficiency of lost-and-found operations by providing a centralized platform for managing item records and facilitating communication between users and administrators.

Inclusions (Key Capabilities): The system shall enable secure user registration, multi-role access control, lost and found item report submissions, item searching, automated match filtering, status notifications, and item recovery tracking.

Exclusions (System Boundaries): The system is strictly an information-tracking and matching platform. It shall not handle monetary rewards, electronic payment processing, physical storage locker integrations, or third-party shipping and courier services.

1.5 Definitions, Acronyms, and Abbreviations

Term	Definition
Admin	A system administrator responsible for managing reports and monitoring system activities.
API	Application Programming Interface — A set of definitions and protocols for building and integrating application software.
LFM / LFMS	Lost and Found Management System — the software application described in this document.
SRS	Software Requirements Specification — a document that describes the intended purpose, requirements, and nature of a software.
UI	User Interface — the visual part of a computer application through which a user interacts with the software.
User	A person who uses the Lost and Found Management System to report or search for items.

1.6 References

1. IEEE Computer Society. IEEE Recommended Practice for Software Requirements Specifications (IEEE Std 830-1998). 1998. <https://standards.ieee.org/ieee/830/1222/> (Normative)
2. Sommerville, I. Software Engineering, 10th Edition. Pearson Education, 2016. (Informative)
3. Object Management Group (OMG). Unified Modeling Language (UML) Specification, Version 2.5. <https://www.omg.org/spec/UML/2.5/> (Normative)
4. Karl E. Wiegers and Joy Beatty. Software Requirements, 3rd Edition. Microsoft Press, 2013. (Normative)
5. Alistair Mavin. Easy Approach to Requirements Syntax (EARS). <https://alistairmavin.com/ears/> (Informative)

2. Overall Description

This section provides background and context influencing the product's requirements.

2.1 Product Perspective

The Lost and Found Management System (LFMS) is a new, self-contained web-based application designed to provide a centralized platform for managing lost and found belongings. The system operates as a standalone product and relies on a centralized database for storing user information, item reports, and recovery records. It allows users to report lost items, report found items, search for matching items, and receive notifications regarding potential matches. Administrators are responsible for managing reports, monitoring system activities, and maintaining data integrity. The system aims to replace manual, paper-based, or fragmented lost-and-found processes, improving efficiency, accessibility, and communication between users and administrators.

System Boundaries: The system interacts directly with users and administrators through standard web browsers and does not depend on external payment, shipping, or storage management services.

Ownership and Support: The software is developed and maintained by the student project team as part of an academic project requirement and does not include formal Service Level Agreements (SLAs) or external support services.

2.2 Product Functions

The LFM System provides several core functions to support the reporting, tracking, and recovery of lost and found items. The main functions of the system include:

- User registration and login.
- Reporting lost items.
- Reporting found items.
- Searching for lost and found items.
- Viewing matching results.
- Receiving notifications when potential matches are identified.
- Updating item status and marking items as returned.
- Managing reports and user activities through an administrator interface.
- Maintaining a centralized database of lost and found records.

2.3 User Classes and Characteristics

The Lost and Found Management System is intended for the following user classes:

1. Users: Users are individuals who access the system to report lost items, report found items, search for items, and receive notifications. They are expected to have basic computer and internet usage skills and should be able to interact with a web-based application.

2. Administrators: Administrators are responsible for managing reports, monitoring system activities, verifying item records, and maintaining system data. Administrators should have sufficient knowledge of system operations and administrative functions to ensure the effective management of the platform.

Note: Users have limited access to their own reports and search functions, while Administrators have privileged access to manage reports, users, and system records.

2.4 Operating Environment

The LFMS shall operate within the following client-server environment:

- **Hardware Platform:** Compatible with desktop PCs, laptops, and mobile devices.
- **Operating Systems:** Compatible with Windows, macOS, Android, and iOS.
- **Software Environment:** Accessible via modern web browsers including Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.
- **Server-Side Environment:** Hosted on a centralized web server utilizing a relational database for data persistence.

2.5 Design and Implementation Constraints

2.5.1 Deployment and Platform Constraints

- **Web-Based Architecture:** The system must be developed as a web-based application that is fully accessible and operational through standard modern web browsers.
- **Environment Compatibility:** Development must be carried out using commonly available software development tools and technologies that are fully compatible with modern operating systems and web environments.

2.5.2 User Interface and Usability Constraints

- **Interface Simplicity:** The user interface must be designed to be simple, intuitive, and user-friendly to accommodate users with varying levels of technical literacy.

2.5.3 Security and Data Protection Constraints

- **Access Control:** The system must ensure the security and confidentiality of user data by enforcing user authentication and access control mechanisms.

2.5.4 Resource and Schedule Constraints

- **Academic Timeline:** Project development and implementation must be completed within the allocated academic schedule.
- **Resource Limits:** The scope of development must remain within the available development resources.

2.6 User Documentation

The system will deliver comprehensive documentation to ensure that both general users and system administrators can successfully interact with and maintain the software. The following user documentation components will be provided:

- **Online Help and FAQ Guide:** An embedded, web-accessible user manual designed for general end-users, guiding them step-by-step through account registration, item report submission, match filtering, and status updates.

- **Administrator Operations Manual:** A technical guide delivered in electronic PDF format tailored for system administrators, detailing data verification workflows, account suspension procedures, report moderation, and database housekeeping tasks.
- **Interactive Tooltips:** UI-level instructional components and inline help icons provided on complex data entry forms to assist users with varying levels of technical literacy.

2.7 Assumptions and Dependencies

This section outlines the assumed environmental factors and external system dependencies that are critical to the project's success. If any of these conditions change or prove incorrect, it may severely impact the system's development schedule or deployment.

2.7.1 Assumptions

- **User Digital Literacy:** End-users possess basic computing and internet browsing skills, allowing them to navigate a standard web form without requiring formal in-person training. Impact if false: increased user friction, poorly filled item reports, and higher reliance on administrative support.
- **Persistent Data Validity:** Users will report accurate metadata (e.g., precise date, time, and location) when uploading items. Impact if false: the automated match-filtering logic will produce low-quality or irrelevant matching results.
- **Continuous Hosting Availability:** The university or local testing server infrastructure will remain operational, stable, and accessible throughout the academic development cycle. Impact if false: the system may become unavailable, preventing access to system functions.

2.7.2 Dependencies

- **Modern Web Browsers:** The system depends on compliant rendering engines within modern client browsers to properly display the user interface and execute asynchronous JavaScript functionality.
- **Relational Database Server:** The application depends on a stable, continuously running relational database management system to manage data persistence and maintain relational integrity across user and report tables.

3. Requirements

This section specifies the verifiable requirements of the software product to enable design and verification. Each requirement uses a unique identifier and the normative keyword “shall”.

3.1 External Interface Requirements

3.1.1 User Interfaces

Req ID	REQ-INT-001
Title	Web-Based User Interface
Statement	The system shall provide a responsive web-based user interface that allows users and administrators to access all system functions through standard web browsers.
Rationale	To ensure easy access and usability across different devices and user groups.
Acceptance Criteria	1. The interface displays correctly on desktop and mobile devices. 2. All major functions are accessible through the web interface.
Verification Method	Demonstration
More Information	Supports the operating environment defined in Section 2.4.
Priority	High

3.1.2 Hardware Interfaces

Req ID	REQ-INT-005
Title	Device Compatibility
Statement	The system shall support interaction through standard hardware devices including keyboards, mice, and touch-screen devices.
Rationale	To ensure compatibility with commonly used user hardware.
Acceptance Criteria	1. Users can successfully interact with the system using keyboard and mouse input. 2. Users can successfully interact with the system using touch-screen devices.
Verification Method	Test
More Information	Related to the operating environment defined in Section 2.4.
Priority	Medium

3.1.3 Software Interfaces

Req ID	REQ-INT-010
Title	Database Integration Interface
Statement	The system shall communicate with a centralized relational database to store, retrieve, and manage user accounts, item reports, and system records.
Rationale	To ensure reliable storage and retrieval of system data.
Acceptance Criteria	1. User and item records are successfully stored in the database.

	2. Stored records can be retrieved correctly when requested.
Verification Method	Test
More Information	Supports database requirements described in Section 2.4.
Priority	High

3.1.4 Communications Interfaces

Req ID	REQ-INT-015
Title	Web Communication Protocol
Statement	The system shall use HTTP and HTTPS protocols for communication between client web browsers and the application server.
Rationale	To ensure reliable and secure data transmission across the network.
Acceptance Criteria	1. Users can access the system through HTTP/HTTPS connections. 2. Data is transmitted successfully between the client and server.
Verification Method	Inspection
More Information	Related to communication requirements and Internet connectivity.
Priority	High

3.2 Functional Requirements

3.2.1 Account Management

3.2.1.1 Description and Priority. This feature enables users to create new accounts and securely access the Lost and Found Management System. Priority: High.

3.2.1.2 Stimulus/Response Sequences. Stimulus: an unregistered visitor enters registration information including name, email address, and password. Response: the system validates the submitted information and creates a new user account if the information is valid.

3.2.1.3 Functional Requirements.

Req ID	REQ-FUNC-001
Title	User Registration
Statement	When an unregistered visitor submits unique registration parameters containing an email address, name, and password, the Lost and Found Management System shall create a new active account record in the database.
Rationale	To establish a unique user identity so users can safely manage their profiles and track items.
Acceptance Criteria	1. User account data is securely saved in the relational database. 2. Duplicate registration using an existing email address is rejected with an error message.
Verification Method	Test
More Information	Relates to data privacy and authentication constraints in Section 2.5.3.
Priority	High

Req ID	REQ-FUNC-002
Title	User Login
Statement	When a registered user submits valid credentials matching an active database record, the Lost and Found Management System shall authenticate the session and open the user dashboard.
Rationale	To secure the system and restrict dashboard functionality to authenticated account owners.
Acceptance Criteria	1. Validated credentials immediately grant system access and display the user dashboard. 2. Incorrect credentials block login access and output a generic “Invalid credentials” error message.
Verification Method	Test
More Information	Relates to role-based system access controls outlined in Section 2.3.
Priority	High

3.2.2 Item Reporting

3.2.2.1 Description and Priority. This feature enables users to submit lost and found item reports. The reports are stored in a centralized database and used for searching and matching purposes. Priority: High.

3.2.2.2 Stimulus/Response Sequences. Stimulus: a user completes and submits a lost item report or a found item report. Response: the system validates the submitted information and stores the report in the database with the appropriate status.

3.2.2.3 Functional Requirements.

Req ID	REQ-FUNC-003
Title	Report Lost Item
Statement	While an authenticated session is active, when a user submits a lost item form detailing the item title, category, date lost, and location lost, the Lost and Found Management System shall log the item status as “Lost” in the database.
Rationale	Allows users to publish details about their missing belongings so that matching algorithms can run.
Acceptance Criteria	1. The lost item report is committed to the database and linked to the active user’s ID. 2. Leaving mandatory fields blank triggers an alert message and stops form submission.
Verification Method	Test
More Information	Primary user capability specified under Product Functions in Section 2.2.
Priority	High

Req ID	REQ-FUNC-004
Title	Report Found Item

Statement	While an authenticated session is active, when a user submits a found item form containing the category, date recovered, and location recovered, the Lost and Found Management System shall log the item status as “Found” in the database.
Rationale	Enables users who discover lost items in public spaces to make them trackable on the platform.
Acceptance Criteria	1. The found item record is saved immediately and becomes searchable in the public database. 2. Submitting an empty form is rejected until core item features are provided.
Verification Method	Test
More Information	Maps directly to the standalone product functionalities in Section 2.1.
Priority	High

3.2.3 Search and Matching

3.2.3.1 Description and Priority. This feature enables users to search for lost and found records, view potential matching results, and receive notifications when matches are identified. Priority: High.

3.2.3.2 Stimulus/Response Sequences. Stimulus: a user performs a search, or the system detects a potential match between lost and found item reports. Response: the system retrieves matching records, displays relevant results, and notifies users when a potential match is found.

3.2.3.3 Functional Requirements.

Req ID	REQ-FUNC-005
Title	Search Lost and Found Items
Statement	When a user enters query parameters into the search bar, the Lost and Found Management System shall filter the centralized database and return all matching item records based on title, location, or category.
Rationale	Gives users the ability to manually look up items to see if their lost belongings have already been reported as found.
Acceptance Criteria	1. The system displays matching database results in a list view. 2. If zero items match the query, the system displays a clear “No matching items found” message.
Verification Method	Test
More Information	Connects closely with search functionality described in Section 2.2.
Priority	Medium

Req ID	REQ-FUNC-006
Title	View Matching Results
Statement	While a user is viewing a submitted report, the Lost and Found Management System shall display a list of potential matching records based on category, date, and location information.
Rationale	Saves users time by collecting likely matches directly inside their active report page.

Acceptance Criteria	1. The system displays a dedicated “Potential Matches” section on the report detail page. 2. Users can open any suggested match to view additional details.
Verification Method	Test
More Information	Works together with the matching functionality described in Section 2.2.
Priority	Medium

Req ID	REQ-FUNC-007
Title	Receive Notifications
Statement	When a new report creates a potential match with an existing report, the Lost and Found Management System shall display a notification to the relevant user.
Rationale	Keeps users informed about potential matches without requiring continuous manual searches.
Acceptance Criteria	1. A notification appears on the user’s dashboard when a match is identified. 2. Selecting the notification opens the related report details.
Verification Method	Demonstration
More Information	Covers communication features highlighted in Product Functions, Section 2.2.
Priority	High

3.2.4 Item Status Management

3.2.4.1 Description and Priority. This feature enables users and administrators to update the status of reported items and mark recovered items as returned. Priority: Medium.

3.2.4.2 Stimulus/Response Sequences. Stimulus: a user or administrator marks an item report as resolved or returned. Response: the system updates the item status in the database and removes the item from active matching and search processes.

3.2.4.3 Functional Requirements.

Req ID	REQ-FUNC-008
Title	Update Item Status and Mark Item as Returned
Statement	When an item owner or administrator marks an item report as resolved, the Lost and Found Management System shall update the status to “Returned”.
Rationale	Keeps the database accurate and prevents recovered items from remaining active.
Acceptance Criteria	1. The item status changes from “Open” to “Returned”. 2. Returned items no longer appear in active search results.
Verification Method	Test
More Information	Linked to item lifecycle management within the system.
Priority	Medium

3.2.5 Administrative Management

3.2.5.1 Description and Priority. This feature provides administrators with tools to manage reports, monitor user activities, and maintain the integrity of system records. Priority: High.

3.2.5.2 Stimulus/Response Sequences. Stimulus: an administrator performs a management action such as editing, verifying, deleting reports, or managing user accounts. Response: the system validates administrator privileges and updates the affected records accordingly.

3.2.5.3 Functional Requirements.

Req ID	REQ-FUNC-009
Title	Manage Reports and User Activities (Administrator)
Statement	While an authorized administrator session is active, the Lost and Found Management System shall provide tools to edit, delete, verify, or manage item reports and user accounts.
Rationale	Provides administrative oversight to maintain data quality and resolve disputes.
Acceptance Criteria	1. Administrative controls are accessible only to administrator accounts. 2. Administrative changes are reflected immediately in the system database.
Verification Method	Test
More Information	Satisfies the administrator privileges defined in Section 2.3.
Priority	High

3.3 Non-Functional Requirements (Quality of Service)

3.3.1 Performance Requirements

Req ID	REQ-NONF-001
Title	System Response Time
Statement	The system shall display search results, user dashboards, and item reports within 3 seconds under normal operating conditions.
Rationale	To ensure a smooth and efficient user experience.
Acceptance Criteria	1. Search results are displayed within 3 seconds. 2. User dashboard pages load within 3 seconds under normal load conditions.
Verification Method	Test
More Information	Applies to REQ-FUNC-005, REQ-FUNC-006, and REQ-FUNC-007.
Priority	High

3.3.2 Safety Requirements

Req ID	REQ-NONF-005
Title	Data Deletion Confirmation
Statement	The system shall require confirmation before permanently deleting any item report or user record.
Rationale	To prevent accidental loss of important information.
Acceptance Criteria	1. A confirmation dialog is displayed before deletion. 2. Records are not deleted unless the user confirms the action.
Verification Method	Demonstration

More Information	Applies to administrative functions.
Priority	High

3.3.3 Security Requirements

Req ID	REQ-NONF-010
Title	Authentication and Access Control
Statement	The system shall require user authentication before granting access to protected functions and shall restrict administrative functions to authorized administrators only.
Rationale	To protect user information and prevent unauthorized access.
Acceptance Criteria	1. Unauthenticated users cannot access protected pages. 2. Administrative functions are accessible only to administrator accounts.
Verification Method	Test
More Information	Supports the security constraints defined in Section 2.5.3.
Priority	High

3.3.4 Software Quality Attributes

3.3.4.1 Reliability

Req ID	REQ-NONF-015
Title	Data Reliability
Statement	The system shall store and retrieve item reports without data corruption or loss during normal operation.
Rationale	To ensure the accuracy and consistency of lost and found records.
Acceptance Criteria	1. Submitted reports remain available after system restart. 2. No report data is lost during normal database operations.
Verification Method	Analysis
More Information	Applies to all report management functions.
Priority	High

3.3.4.2 Availability

Req ID	REQ-NONF-020
Title	System Availability
Statement	The system shall be available to users at least 95% of the time, excluding scheduled maintenance periods.
Rationale	To ensure users can access the platform whenever required.
Acceptance Criteria	1. System uptime is maintained at or above 95%. 2. Scheduled maintenance is announced in advance.

Verification Method	Analysis
More Information	Applies to all system services.
Priority	Medium

3.3.5 Business Rules

Req ID	REQ-NONF-025
Title	Report Ownership Rule
Statement	The system shall allow users to modify only their own reports, while administrators shall have permission to manage all reports and user accounts.
Rationale	To preserve data ownership and maintain administrative control.
Acceptance Criteria	1. Users cannot edit reports created by other users. 2. Administrators can manage all reports and accounts.
Verification Method	Inspection
More Information	Supports user roles defined in Section 2.3.
Priority	High

3.4 Other Requirements

Req ID	REQ-OTH-001
Title	Database Backup
Statement	The system shall maintain periodic backups of database records to prevent data loss.
Rationale	To ensure data can be recovered in case of system failure.
Acceptance Criteria	1. Database backups are created successfully. 2. Backup files can be restored when required.
Verification Method	Inspection
More Information	Applies to all user and item records.
Priority	Medium

4. Verification

This section describes how each requirement will be verified to provide objective evidence of compliance. The verification methods below are consistent with the methods declared in Section 3.

Requirement ID	Verification Method	Test / Artifact Link	Status	Evidence
REQ-INT-001	Demonstration	Test Case TC-001	Planned	UI Screenshots
REQ-INT-005	Test	Test Case TC-002	Planned	Device Testing Report
REQ-INT-010	Test	Test Case TC-003	Planned	Database Test Results
REQ-INT-015	Inspection	Test Case TC-004	Planned	Network Configuration
REQ-FUNC-001	Test	Test Case TC-005	Planned	Registration Test Results
REQ-FUNC-002	Test	Test Case TC-006	Planned	Login Test Results
REQ-FUNC-003	Test	Test Case TC-007	Planned	Lost Item Submission Record
REQ-FUNC-004	Test	Test Case TC-008	Planned	Found Item Submission Record
REQ-FUNC-005	Test	Test Case TC-009	Planned	Search Function Results
REQ-FUNC-006	Test	Test Case TC-010	Planned	Matching Results Screenshots
REQ-FUNC-007	Demonstration	Test Case TC-011	Planned	Notification Screenshots
REQ-FUNC-008	Test	Test Case TC-012	Planned	Status Update Record
REQ-FUNC-009	Test	Test Case TC-013	Planned	Admin Dashboard Screenshots
REQ-NONF-001	Test	Test Case TC-014	Planned	Performance Test Report
REQ-NONF-005	Demonstration	Test Case TC-015	Planned	Safety Confirmation Screenshots
REQ-NONF-010	Test	Test Case TC-016	Planned	Security Review Report
REQ-NONF-015	Analysis	Test Case TC-017	Planned	Reliability Analysis Report
REQ-NONF-020	Analysis	Test Case TC-018	Planned	Availability Report
REQ-NONF-025	Inspection	Test Case TC-019	Planned	Access Control Verification
REQ-OTH-001	Inspection	Test Case TC-020	Planned	Backup & Restore Log

Appendix A: Glossary

Term / Acronym	Definition
Administrator	A privileged system user responsible for verifying claims, managing accounts, and maintaining data moderation.
End-User	A registered user who reports lost or found items, searches records, and receives notifications.
API	Application Programming Interface. A set of protocols that allows different software components to communicate.
Bcrypt	A secure cryptographic password-hashing function used to protect user passwords at rest.
EARS	Easy Approach to Requirements Syntax. A structured natural-language template used to write clear, testable requirements.
HTTPS	Hypertext Transfer Protocol Secure. The encrypted protocol used to secure communication between client and server.
LFMS	Lost and Found Management System. The web-based application specified in this document.
RBAC	Role-Based Access Control. A method of restricting access based on user roles and permissions.
SRS	Software Requirements Specification. The document that defines the functional and non-functional requirements of the system.

Appendix B: Analysis Models

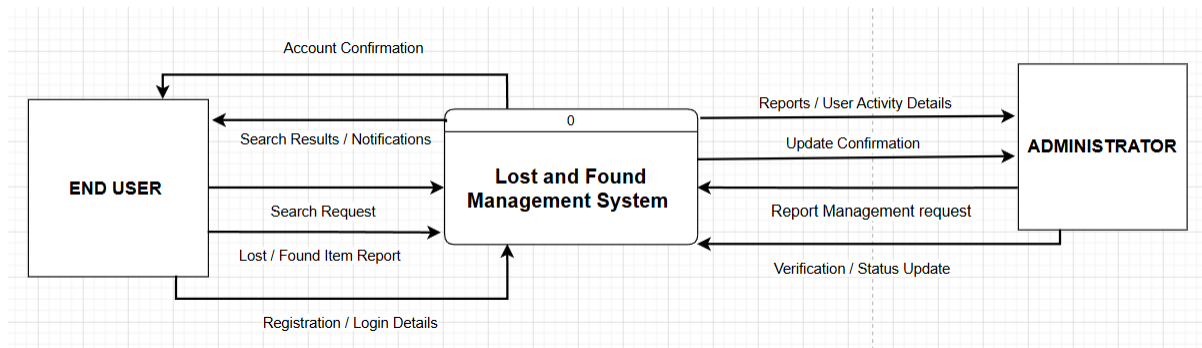


Figure 1 : Context Diagram of the Lost and Found Management System

This context diagram illustrates the overall interaction between the Lost and Found Management System, end users, and administrators, including the exchange of registration details, item reports, search requests, notifications, and administrative actions.

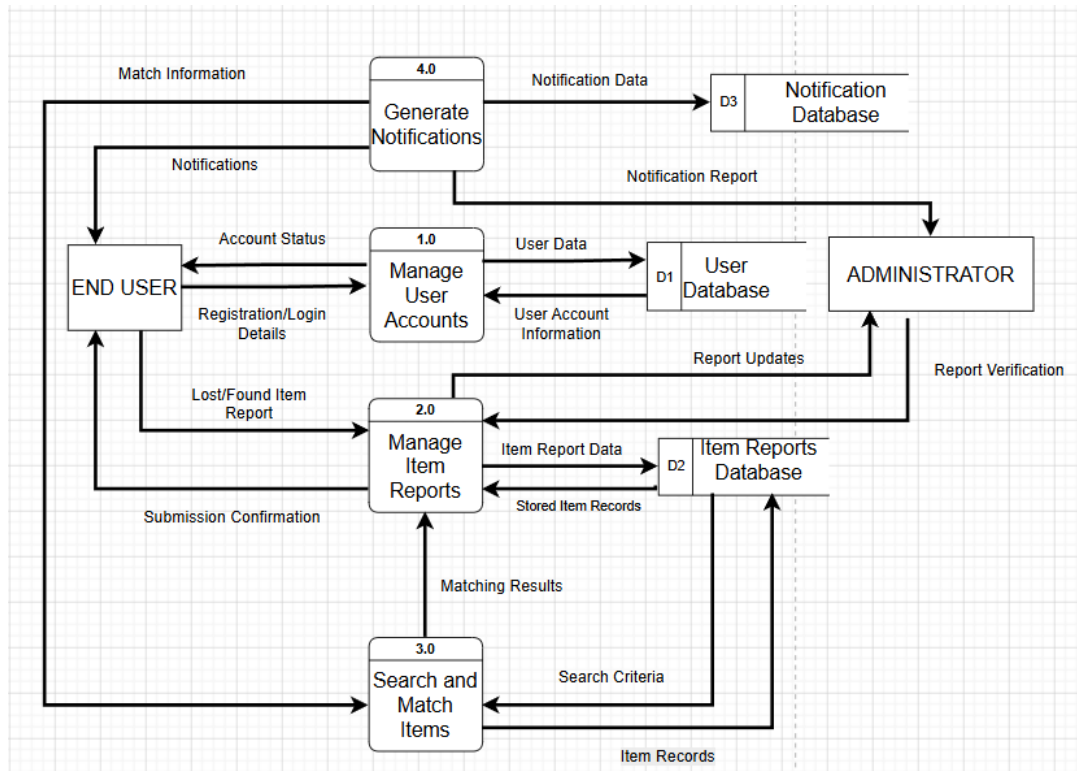


Figure 2 : Data Flow Diagram (Level 0) of the Lost and Found Management System

This Level-0 Data Flow Diagram illustrates the major system processes, data stores, external entities, and data flows involved in user account management, item reporting, item matching, and notification generation.

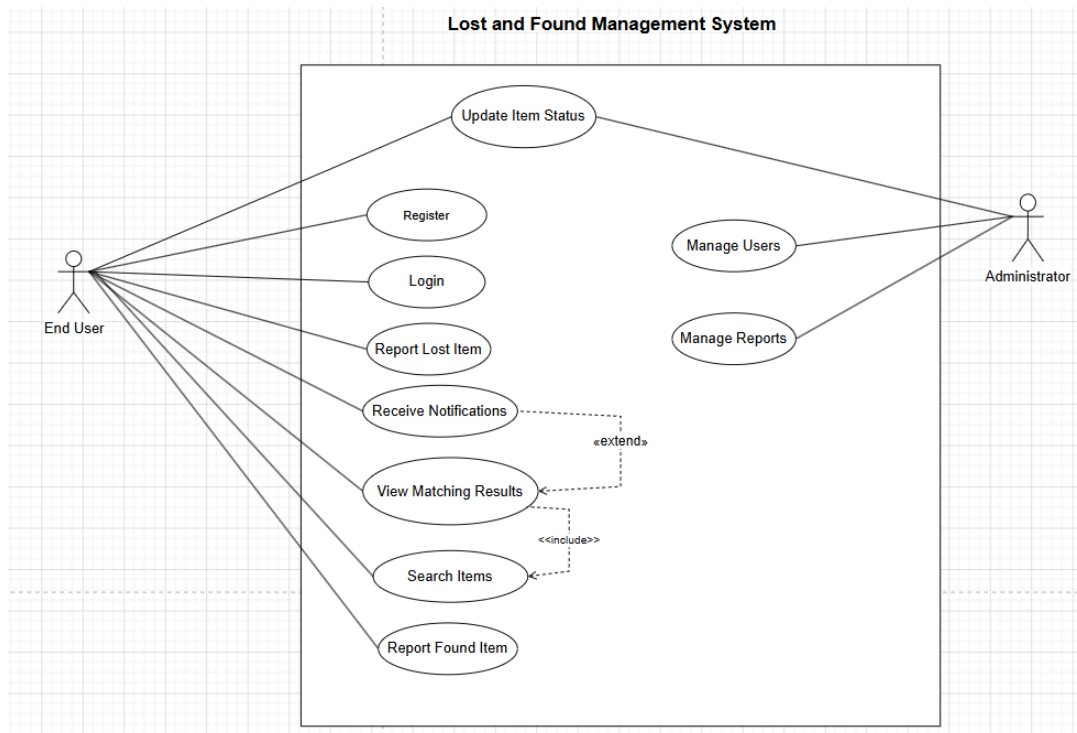


Figure 3 : Use Case Diagram of the Lost and Found Management System

This use case diagram presents the primary interactions between end users and administrators with the system, including registration, login, item reporting, searching, notification handling, status updates, and administrative management functions.

Figure 4 : Activity Diagram for Reporting a Lost Item

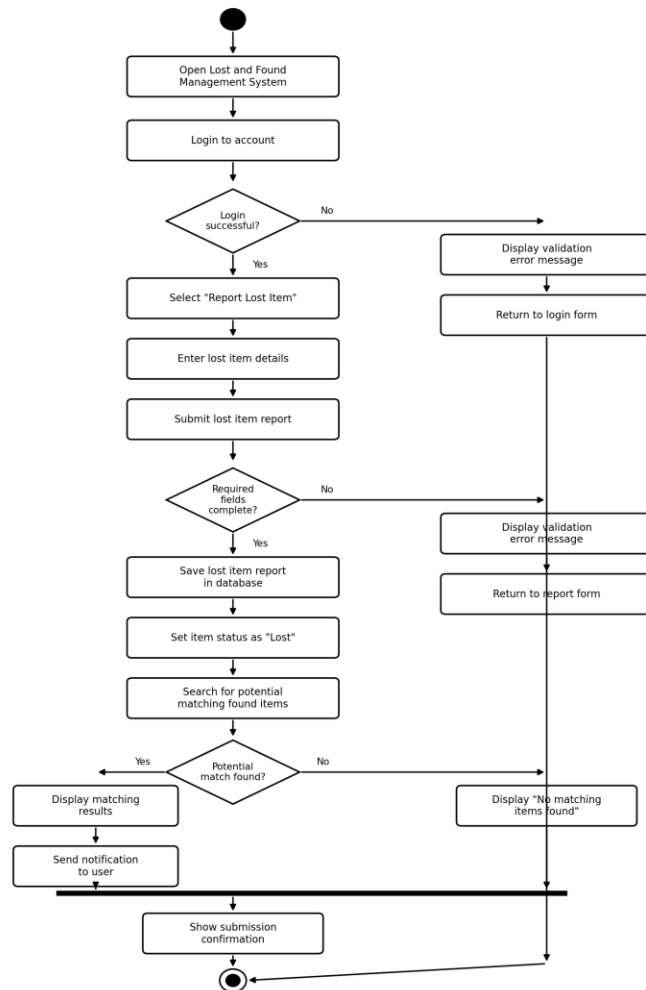
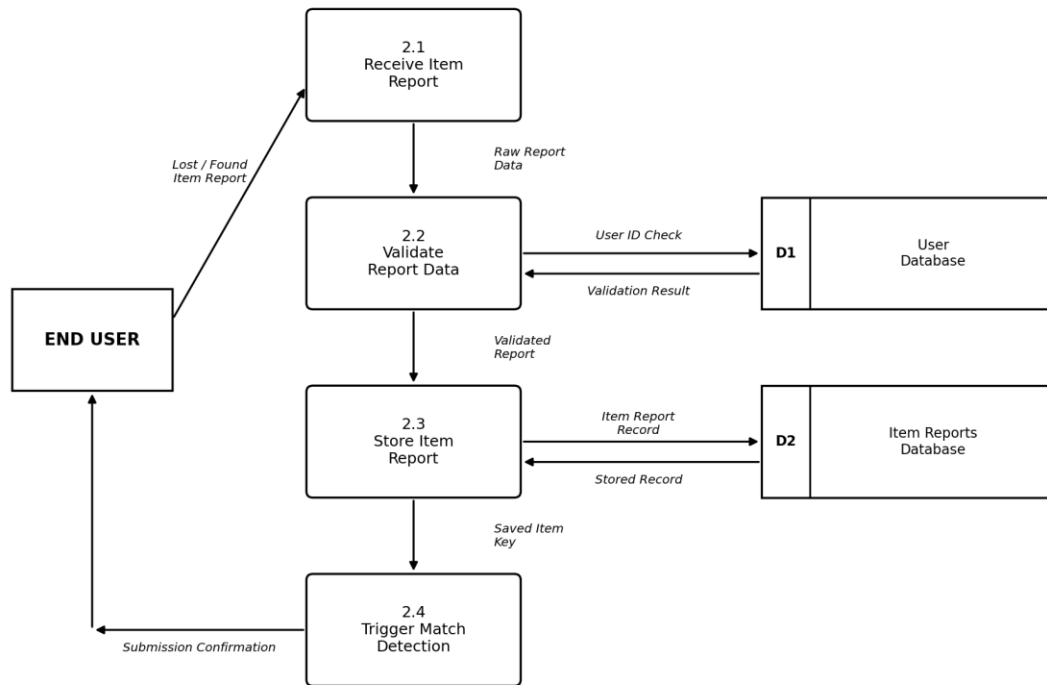


Figure 4 : Activity Diagram for Reporting a Lost Item

This activity diagram illustrates the workflow of reporting a lost item, including user authentication, form submission, validation, database storage, matching detection, notification generation, and confirmation display. The fork/join synchronization bars merge the parallel display and notification flows before the final state.

Figure 5 : Level-1 Data Flow Diagram (Process 2.0 — Manage Item Reports)**Figure 5 : Level-1 Data Flow Diagram (Process 2.0 — Manage Item Reports)**

This Level-1 Data Flow Diagram decomposes Process 2.0 (Manage Item Reports) from the Level-0 DFD into four sub-processes: 2.1 Receive Item Report, 2.2 Validate Report Data, 2.3 Store Item Report, and 2.4 Trigger Match Detection. It shows the detailed data flows between these sub-processes, the End User, and the User and Item Reports databases.

Figure 6 : Class Diagram of the Lost and Found Management System

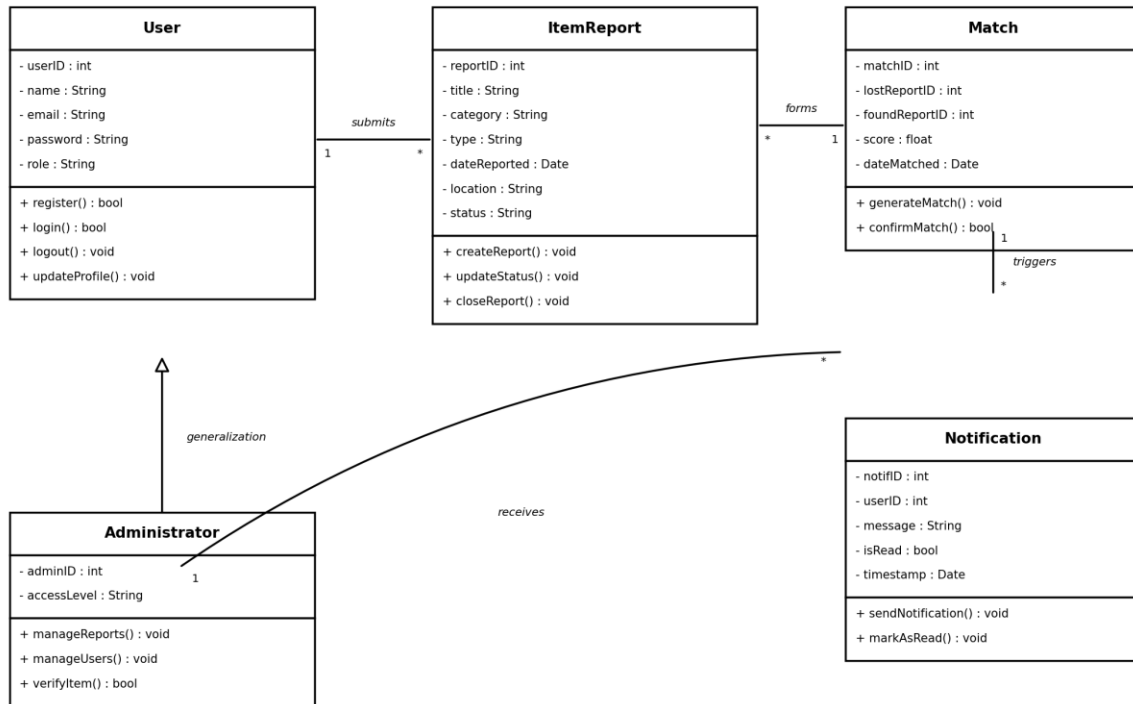


Figure 6 : Class Diagram of the Lost and Found Management System

This class diagram presents the static structure of the system. The Administrator class generalizes (inherits from) the User class. A User submits many ItemReports; matched lost and found reports form a Match, which in turn triggers Notifications delivered to the relevant User. Each class lists its key attributes and operations together with their visibility.

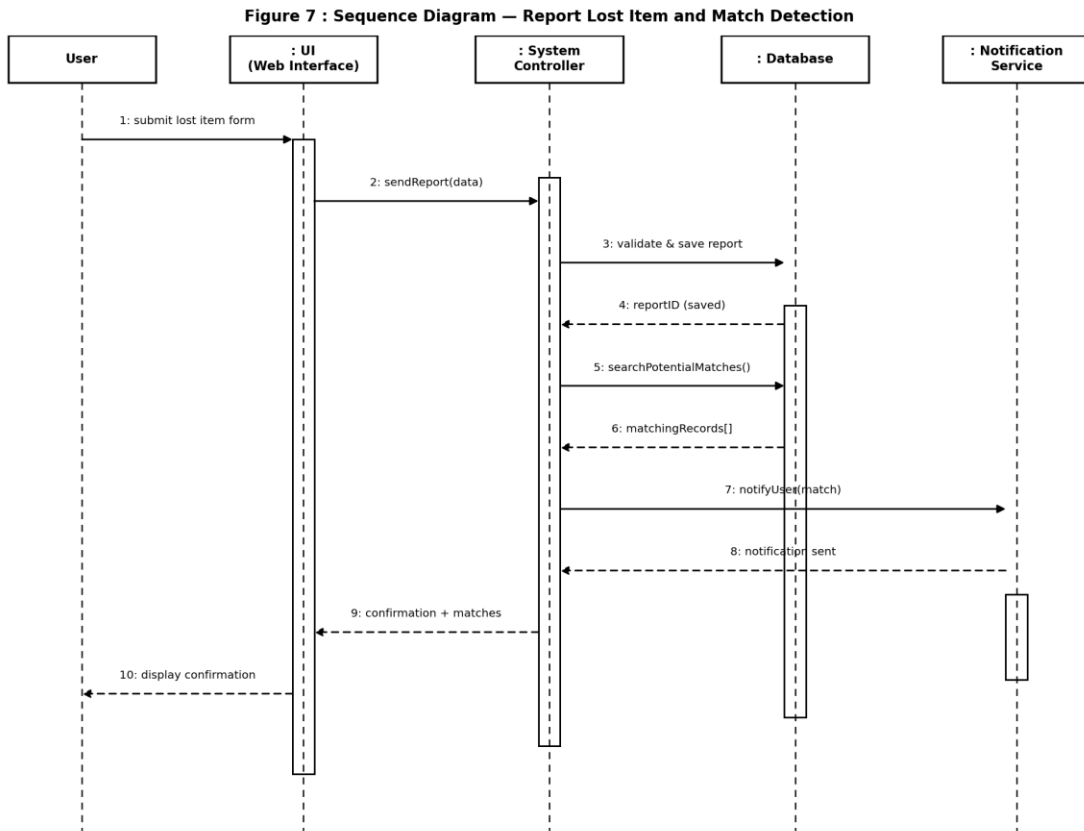


Figure 7 : Sequence Diagram — Report Lost Item and Match Detection

This sequence diagram depicts the dynamic interaction for the “Report Lost Item” scenario. The User submits a lost item form through the Web Interface, which forwards the data to the System Controller. The controller validates and saves the report in the Database, searches for potential matches, and, when a match is found, invokes the Notification Service before returning the confirmation and matching results to the User.

Appendix C: To Be Determined List

No outstanding To Be Determined (TBD) items have been identified at the time of writing this Software Requirements Specification document.